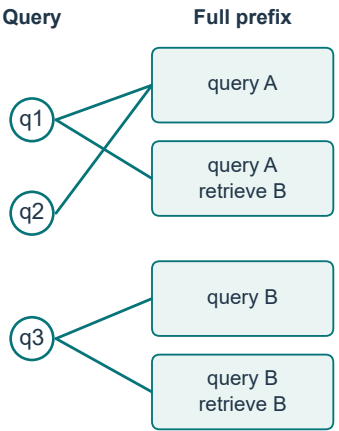
a Query–prefix cache

An edge means a restorable execution state.

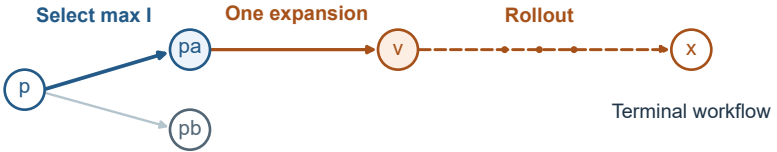


Same query + full prefix

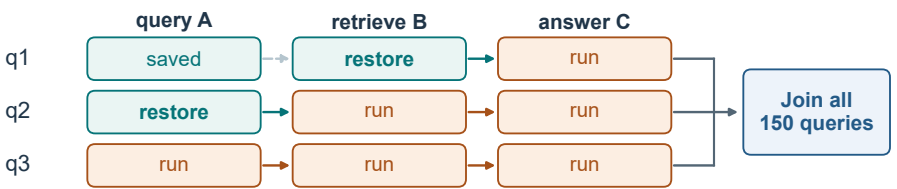
b AgentUCT: price a complete reward

$$I_t(p,a) = Q_t(pa) + c\sqrt{\log(N_t(p)+1)/N_t(pa)} - \beta\,\widehat{b}_t(p,a)/C_{\text{ref}}$$

Price all remaining work under 16 sampled legal completions.



c WTB restores state and executes the new suffix



AgentUCT: one mean F1 → raw backup; update prices / continuation counts.

Expanded ancestors: N ← N + 1, S ← S + F1, Q ← S / N.

Illustrative cache state and tree depth. Three of 150 training queries shown; RAG action labels are abbreviated.